Chapter 8

More Dimensions Than Meet the Eye

Einstein resolved two of the major scientific conflicts of the past hundred years through special and then general relativity. Although the initial problems that motivated his work did not portend the outcome, each of these resolutions completely transformed our understanding of space and time. String theory resolves the third major scientific conflict of the past century and, in a manner that even Einstein would likely have found remarkable, it requires that we subject our conceptions of space and time to yet another radical revision. String theory so thoroughly shakes the foundations of modern physics that even the generally accepted number of dimensions in our universe—something so basic that you might think it beyond questioning—is dramatically and convincingly overthrown.

The Illusion of the Familiar

Experience informs intuition. But it does more than that: Experience sets the frame within which we analyze and interpret what we perceive. You would no doubt expect, for instance, that the "wild child" raised by a pack of wolves would interpret the world from a perspective that differs substantially from your own. Even less extreme comparisons, such as those between people raised in very different cultural traditions, serve to underscore the degree to which our experiences determine our interpretive mindset.

Yet there are certain things that we all experience. And it is often the beliefs and expectations that follow from these universal experiences that can be the hardest to identify and the most difficult to challenge. A simple but profound example is the following. If you were to get up from reading this book, you could move in three independent directions—that is, through three independent, spatial dimensions. Absolutely any path you follow—regardless of how complicated—results from some combination of motion through what we might call the "left-right dimension," the "back-forth dimension," and the "up-down dimension." Every time you take a step you implicitly make three separate choices that determine how you move through these three dimensions.